

Lecture 15: Balance of Payments and Exchange Rates

Maryam Alhalboni PhD.
Maryam.Alhalboni @york.ac.uk

"Exchange rates are like the weather; they can change quickly and unpredictably, but understanding the underlying patterns helps us anticipate their movements."



Lecture overview

- The Balance of Payments
- Exchange Rates
- Exchange Rates and the Balance of Payments
- Fixed versus Floating Exchange Rates

The Balance of Payments

The Balance of Payments Account

Balance of payments (BoP) records all monetary flows between residents and the rest of the world.

- Credits (inflows): money received from abroad.
- Debits (outflows): money sent abroad.
- Three components: Current account, Capital account, Financial account.

The current account

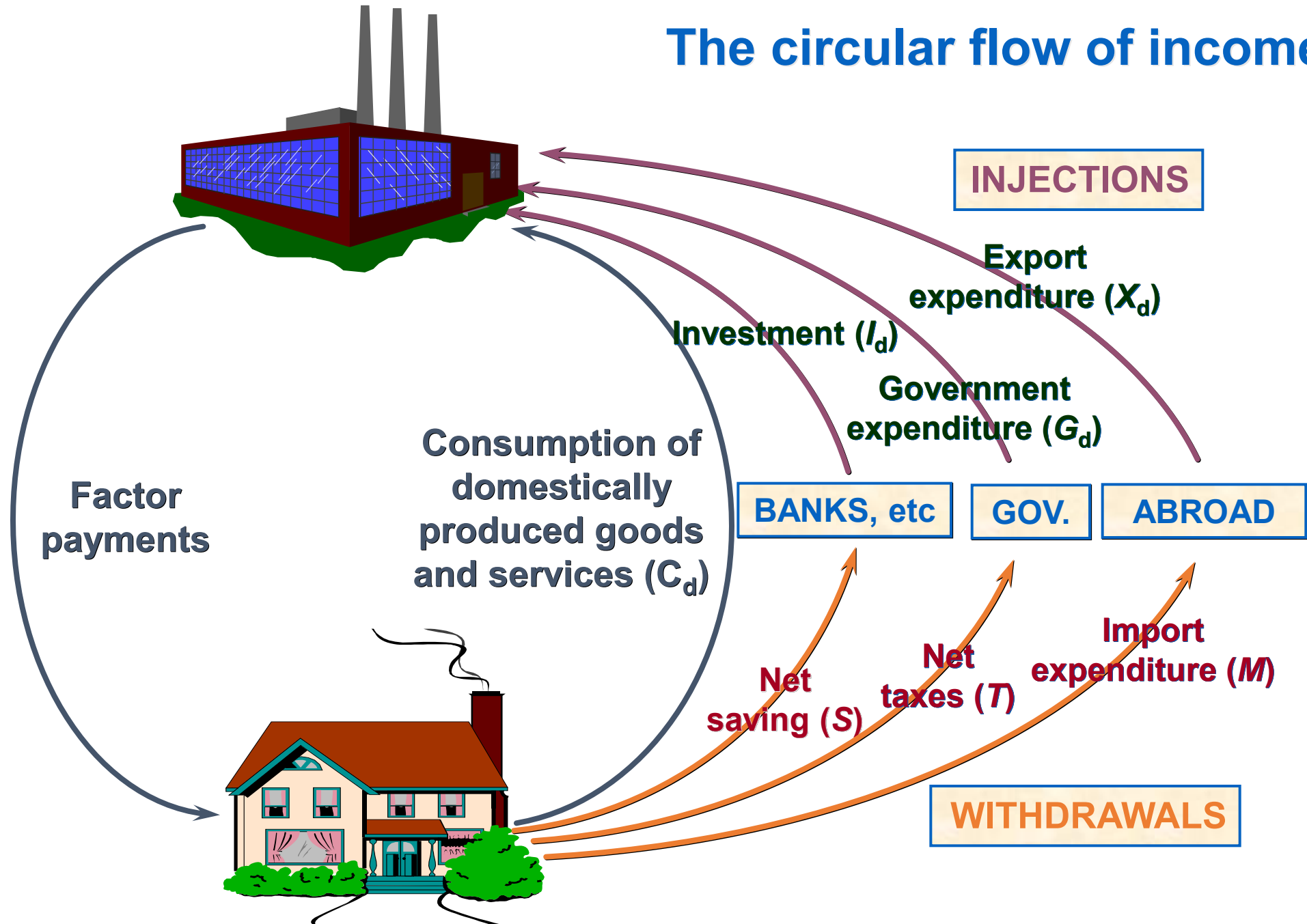
- Trade in goods: Exports (credits) vs imports (debits).
- Trade in services: Tourism, insurance, etc.

balance of trade in goods and services

- Trade deficits = leakages from circular flow (reduced demand) vs Trade surpluses = injections (increased demand)
- **income flows**
- **current transfers** of money
- Current account balance = total of all the above.

Surplus = credits > debits; Deficit = debits > credits.

The circular flow of income



UK balance of payments

	2021		Average 1987–2021 as % of GDP
	£m	% of GDP	
CURRENT ACCOUNT			
Balance on trade in goods	–156 066	–6.7	–4.5
Balance on trade in services	126 959	5.5	3.2
<i>Balance of trade</i>	–29 107	–1.3	–1.3
Income balance	–11 912	–0.5	–0.4
Net current transfers	–18 931	–0.8	–0.9
<i>Current account balance</i>	–59 950	–2.6	–2.6

Source: Based on data from [Balance of Payments time series](#) and series [YBHA](#) (Office for National Statistics)

The capital account

- Records flows related to:
 - Fixed assets (land, patents, trademarks).
 - Migrants' transfers.
 - Government debt forgiveness.
- Usually small compared to other accounts.

UK balance of payments

	2021		Average 1987–2021 as % of GDP
	£m	% of GDP	
CURRENT ACCOUNT			
Balance on trade in goods	–156 066	–6.7	–4.5
Balance on trade in services	126 959	5.5	3.2
<i>Balance of trade</i>	–29 107	–1.3	–1.3
Income balance	–11 912	–0.5	–0.4
Net current transfers	–18 931	–0.8	–0.9
<i>Current account balance</i>	–59 950	–2.6	–2.6
CAPITAL ACCOUNT			
<i>Capital account balance</i>	–2 697	–0.1	0.0

Source: Based on data from [Balance of Payments time series](#) and series [YBHA](#) (Office for National Statistics)

The financial account

- Records cross-border changes in asset ownership.
- **Direct investment:** Long-term interest in foreign businesses.
- **Portfolio investment:** Buying shares/debt without control.
- **Other financial flows:** Short-term monetary movements (e.g., bank deposits, loans).
 - Driven by interest rate differences and exchange rate expectations.
- **Reserves:**
 - Selling reserves to buy sterling = credit.
 - Building reserves = debit.

Net recording: Only net changes are recorded.

The Balance of Payments



School for Business
and Society

Overall BoP must balance: if not, exchange rates adjust or government intervenes.

Net errors and omissions

- Statistical discrepancies are inevitable.
- A net errors and omissions item is included to ensure BoP balances.
- Causes:
 - Data from multiple sources.
 - Reporting delays or missing data.

UK balance of payments

	2021		Average 1987–2021 as % of GDP
	£m	% of GDP	
CURRENT ACCOUNT			
Balance on trade in goods	–156 066	–6.7	–4.5
Balance on trade in services	126 959	5.5	3.2
<i>Balance of trade</i>	–29 107	–1.3	–1.3
Income balance	–11 912	–0.5	–0.4
Net current transfers	–18 931	–0.8	–0.9
<i>Current account balance</i>	–59 950	–2.6	–2.6
CAPITAL ACCOUNT			
<i>Capital account balance</i>	–2 697	–0.1	0.0
FINANCIAL ACCOUNT			
Net direct investment	–58 296	–2.5	–0.4
Portfolio investment balance	254 008	11.0	3.2
Other investment balance	–162 370	–7.0	0.1
Balance of financial derivatives	28 647	1.2	0.0
Reserve assets	–17 701	–0.8	–0.2
<i>Financial account balance</i>	44 288	1.9	2.6

Source: Based on data from [Balance of Payments time series](#) and series [YBHA](#) (Office for National Statistics)

Q If there is a current account deficit of £1bn, then:

- A. there must be a surplus of £1bn on trade in services.
- B. there must be an equivalent deficit on the capital plus financial accounts.
- C. there must be a net errors and omissions item of +£1bn.
- D.

 the overall capital plus financial accounts (including net errors and omissions) must be in surplus by £1bn.
- E. the financial account must be +£1bn.

If there is a current account deficit of £1bn, then:

Option D is correct because a current account deficit must be balanced by an equivalent surplus in the capital and financial accounts (including net errors and omissions) to maintain the overall balance of payments. This ensures the total of the current account and financial account is zero, with any discrepancies accounted for in the net errors and omissions item.

Exchange Rates

Exchange Rates

- Exchange rate = the price of one currency in terms of another.
- Used by individuals and firms for travel, trade, and investment.
- Example: £1 = €1.15 or £1 = \$1.35.
- Foreign exchange markets determine daily exchange rates.
- Dealers aim to balance currency supply and demand to avoid losses.
- Exchange rates change constantly due to market forces and dealer adjustments.

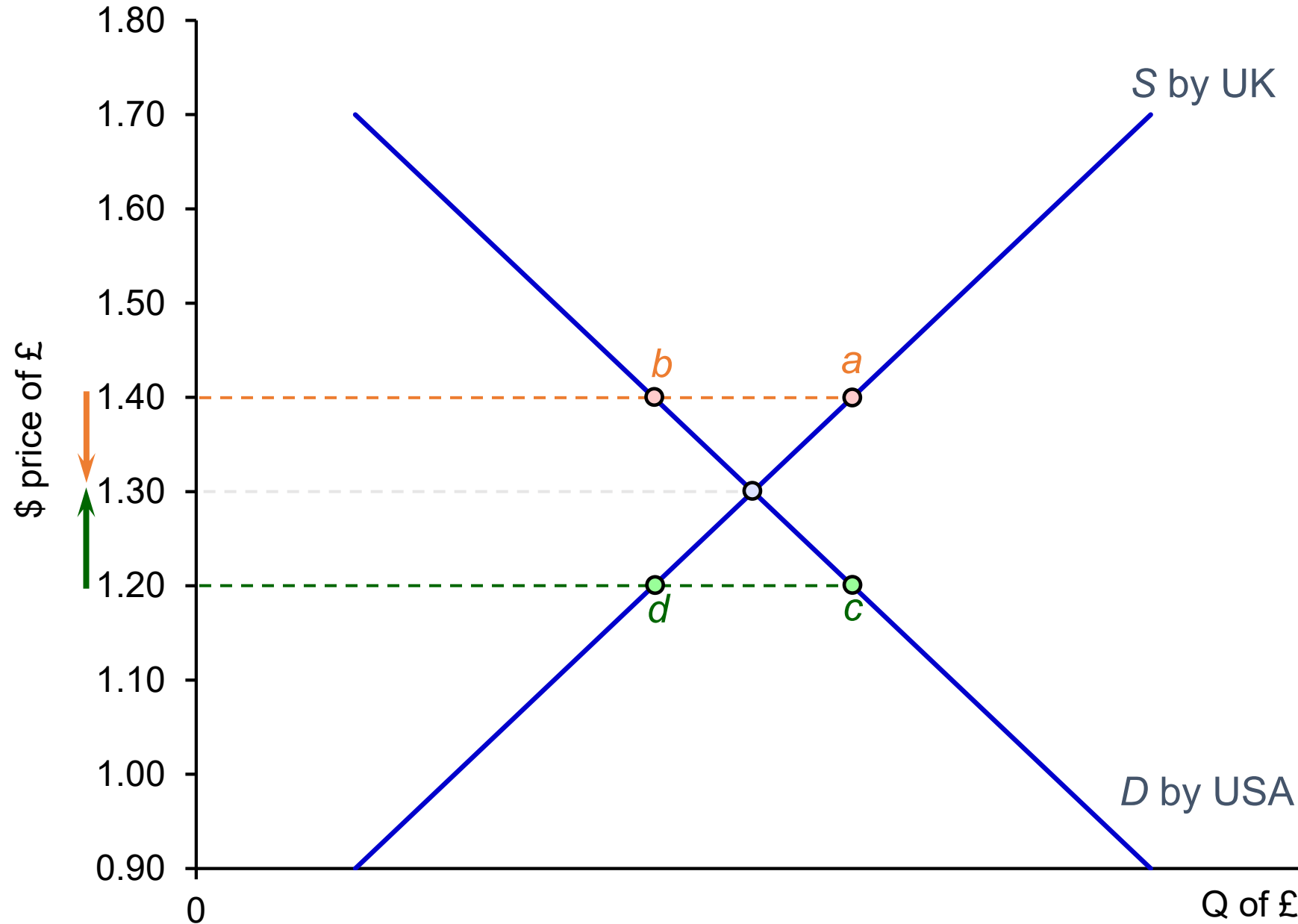
Determining the Exchange Rate in a Free Market

- Exchange rates determined by demand and supply for a currency.
- UK importers sell pounds to buy foreign goods → increases pound supply.
- US buyers of UK goods demand pounds → increases pound demand.
- Supply curve of pounds = upward sloping; demand curve = downward sloping.

Determining the Exchange Rate in a Free Market

- Equilibrium rate occurs where demand equals supply (e.g., £1 = \$1.30).
- Excess supply (e.g., at \$1.40) → rate falls;
- excess demand (e.g., at \$1.20) → rate rises.
- Rates adjust quickly due to competition among dealers.

Determination of the rate of exchange



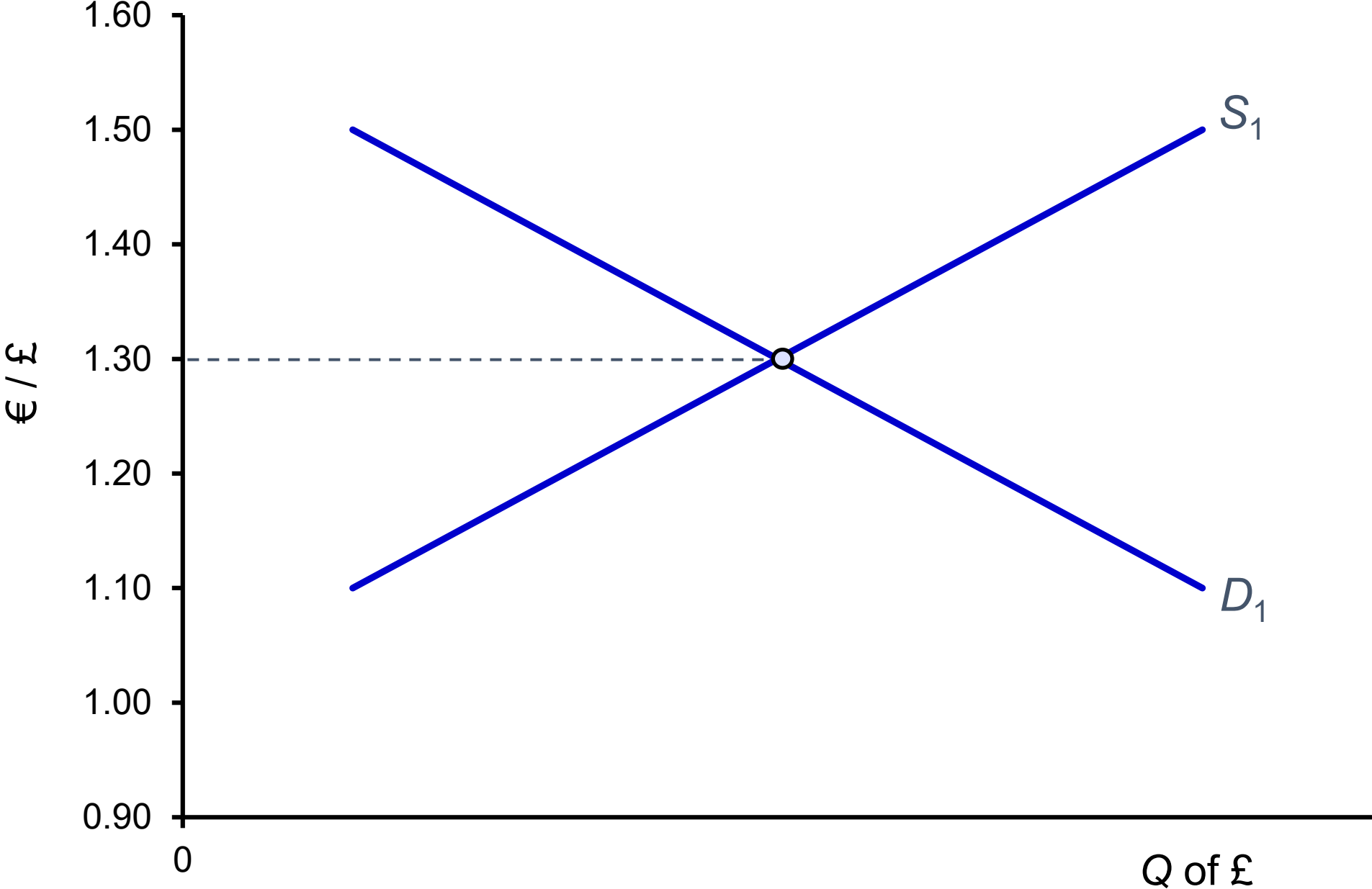
Excess supply of pounds leads to a depreciation.

Shortage of pounds leads to an appreciation.

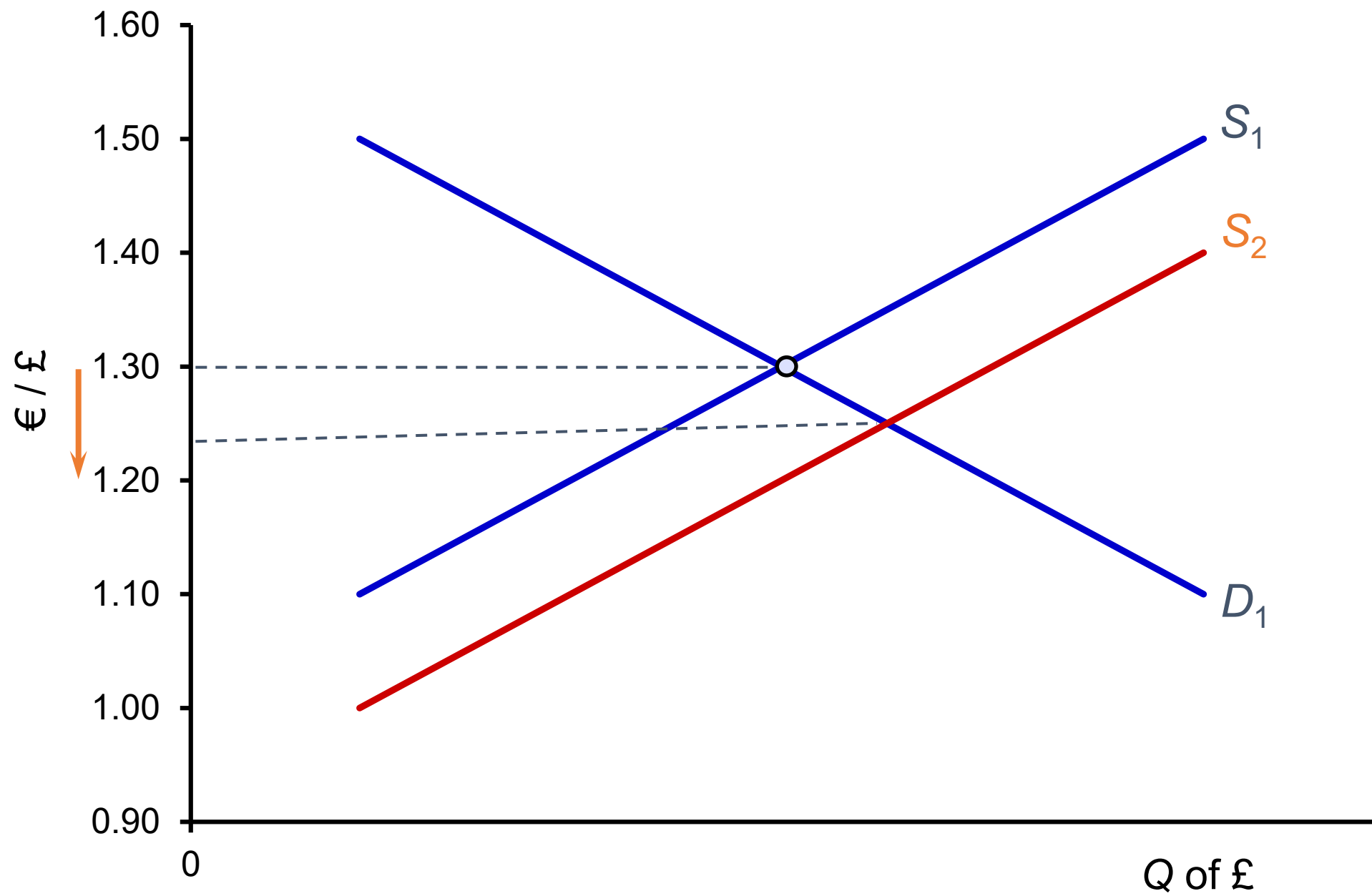
Shifts in Currency Demand and Supply

- Exchange rates change when demand or supply curves shift.
- Examples a euro/sterling.
- Depreciation = fall in exchange rate (e.g., €1.30 → €1.20).
- Appreciation = rise in exchange rate.

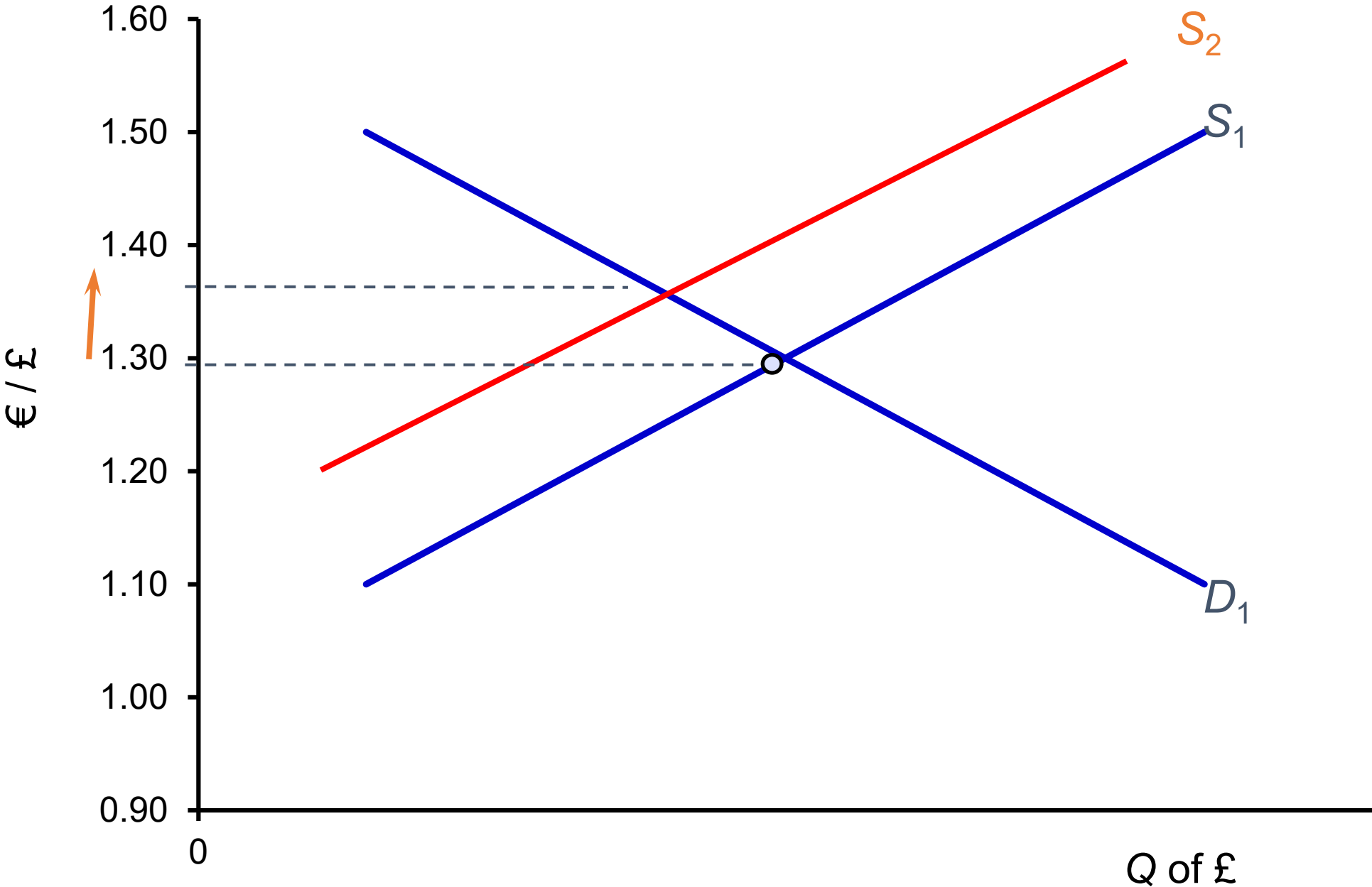
Floating exchange rates:
movement to a new equilibrium



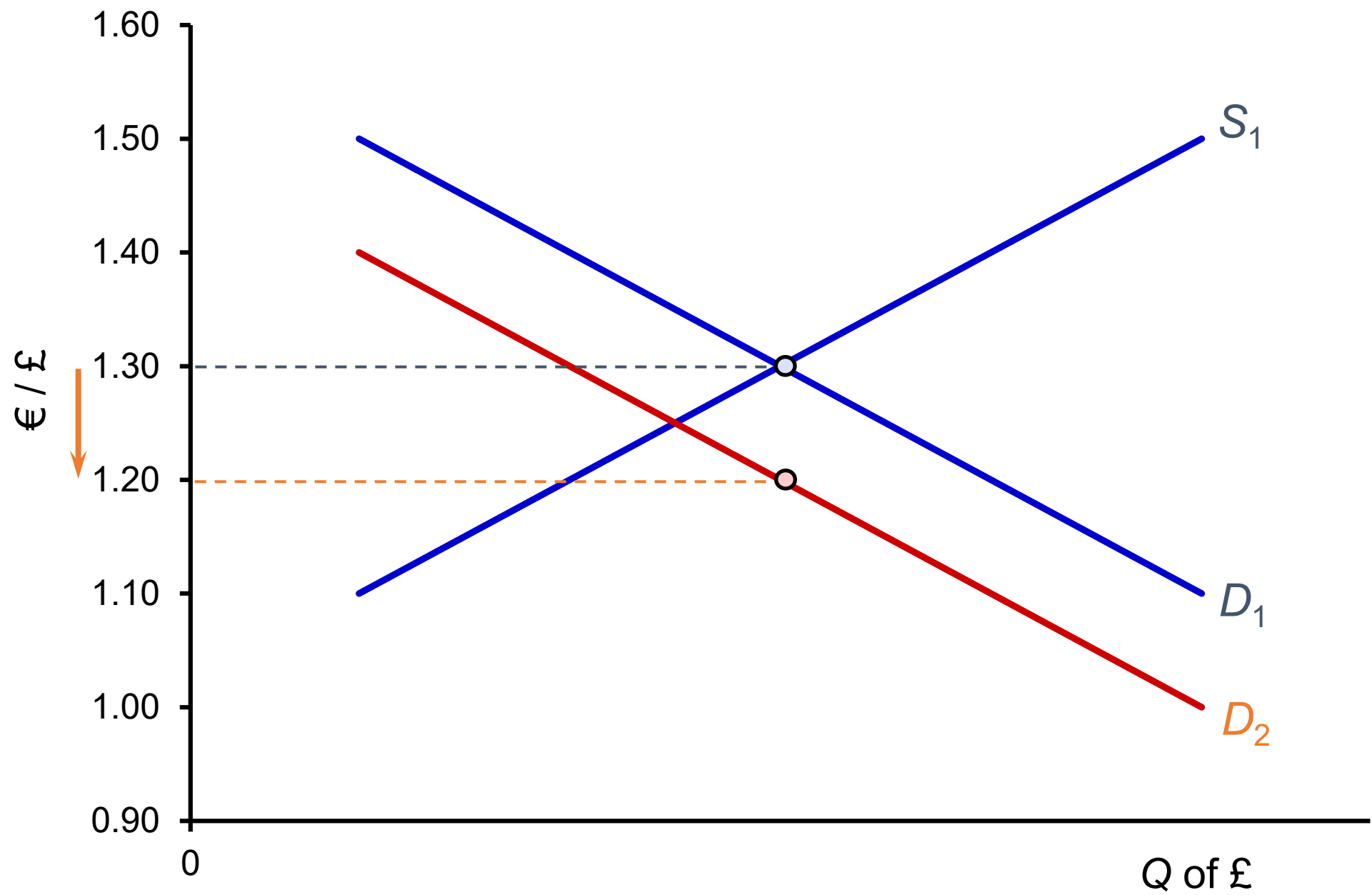
Floating exchange rates:
movement to a new equilibrium



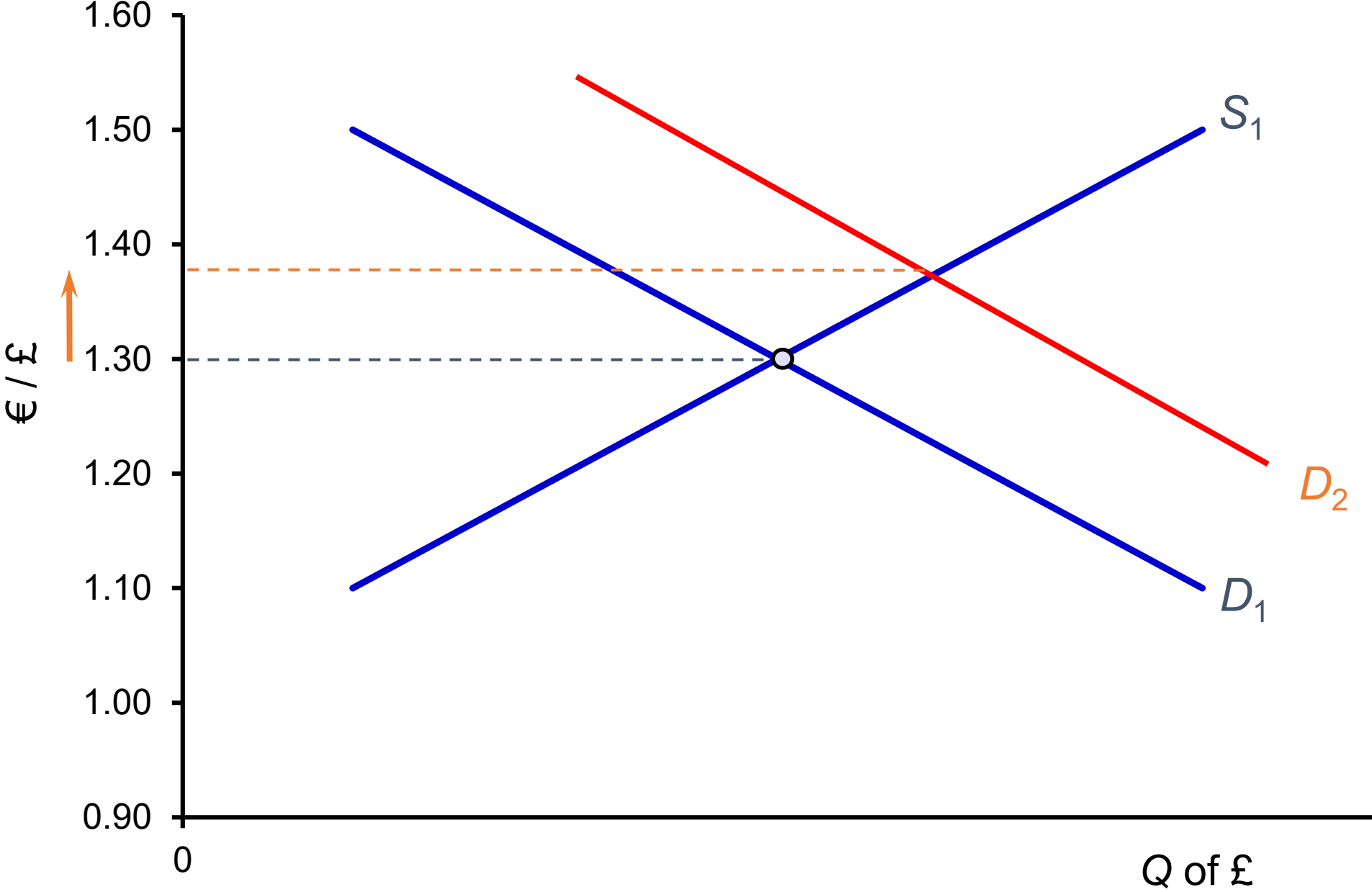
Floating exchange rates:
movement to a new equilibrium



Floating exchange rates:
movement to a new equilibrium



Floating exchange rates:
movement to a new equilibrium



Q Only one of the following flows represents a *demand* for sterling.
Which one?

- A. Imports of goods and services into the UK
- B. UK investment abroad
- C. Short-term financial outflows from the UK
- D.

 Profit earned from UK investment abroad
- E. Overseas aid by the UK government

Only one of the following flows represents a *demand* for sterling.

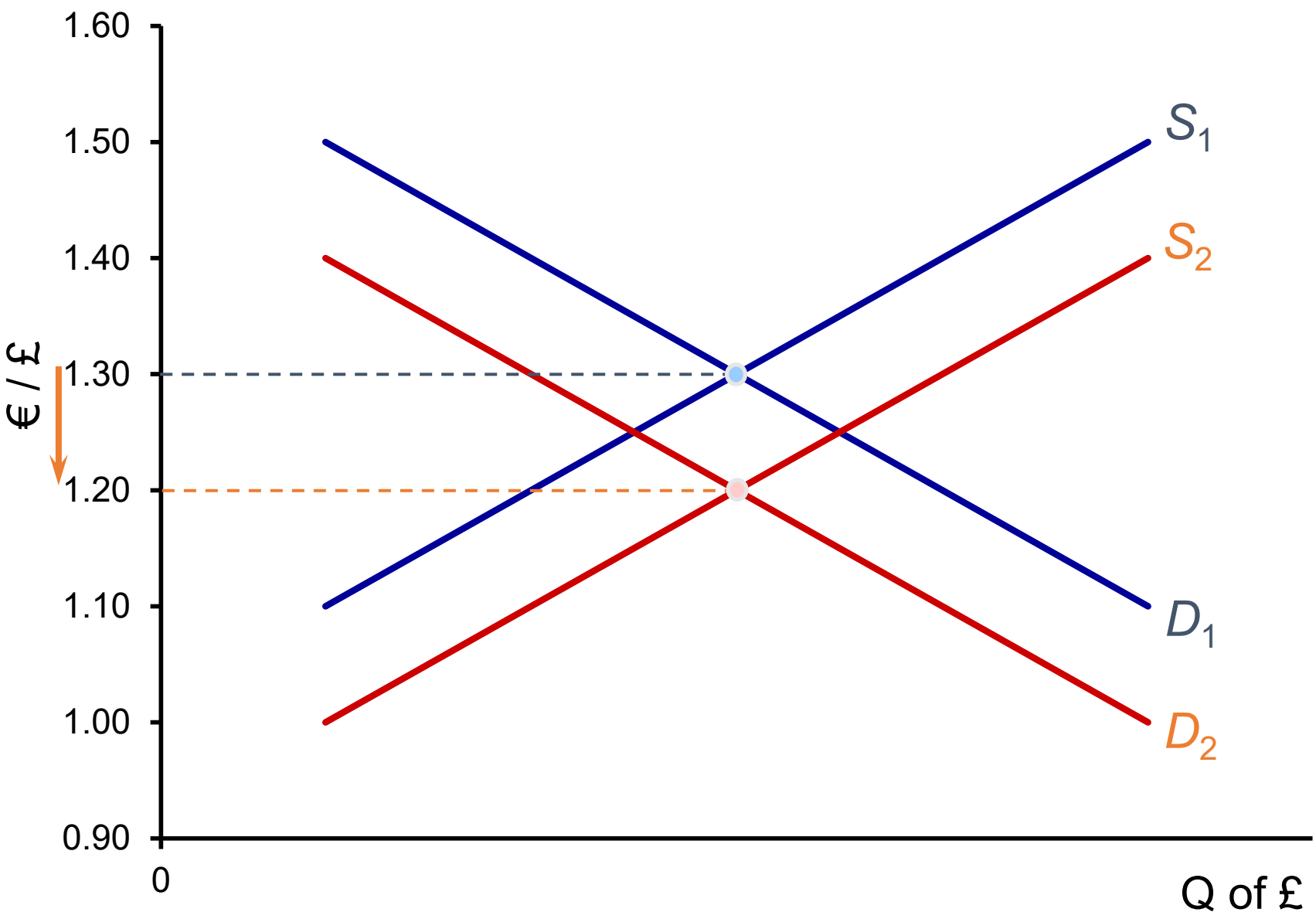
- **A:** Does not create demand for sterling; paid in foreign currencies.
- **B:** Requires converting sterling into foreign currencies, not creating demand for sterling.
- **C:** Involves exchanging sterling for foreign currencies, not creating demand for sterling.
- **E:** Involves exchanging sterling for foreign currencies, not creating demand for sterling.

D is correct because when profit is earned from UK investment abroad and brought back to the UK, there is a demand for sterling as the foreign earnings are converted into pounds.

Causes of Depreciation

- Lower UK interest rates → less attractive to savers → less demand, more supply of sterling.
- Higher UK inflation → exports less competitive → demand falls, imports rise → supply increases.
- Higher UK incomes → more imports → increased supply of sterling.
- Better investment prospects abroad → capital outflows → demand falls, supply rises.
- Speculation of falling exchange rate → increased selling of sterling → supply rises.

Causes of a depreciation (assume no changes abroad)



- fall in interest rate
- rise in inflation rate
- rise in aggregate demand
- inward investment less attractive
- speculation against depreciation

Exchange Rates and the Balance of Payments

Exchange Rates and the Balance of Payments (No Intervention)



School for Business
and Society

Exchange rates and the balance of payments: **without** government or central bank intervention

- a floating exchange rate ensures balance of payments equilibrium.
- automatic balancing of overall balance of payments
- Current, capital and financial accounts may not separately balance

Exchange Rates and the Balance of Payments (With Intervention)



School for Business
and Society

Exchange rates and the balance of payments: **with** government or central bank intervention

- **Short-term intervention:**
 - Use reserves to buy pounds.
 - Borrow foreign currency to buy pounds.
 - Raise interest rates to attract capital.
- **Long-term fixed exchange rate maintenance:**
 - **Contractionary policy:** cut demand via tax hikes or interest rate rises.
 - Lowers imports and inflation → supports exchange rate.
 - **Supply-side policies:** boost competitiveness via training, R&D.
 - **Controls on imports/foreign exchange:** e.g., restrict access to foreign currency or impose tariffs.

Fixed versus Floating Exchange Rates

Fixed versus Floating Exchange Rates

Advantages of fixed exchange rates

- **Certainty:** Reduces risk in international trade and investment.
- **Stability:** Helps businesses plan with less risk of currency fluctuations.
- **Reduced speculation:** If fixed credibly, there's no incentive for currency speculation.
- **Macro discipline:** Forces governments to maintain stable inflation and avoid reckless economic policies.
- **International confidence:** Encourages stable investment climates.

Fixed versus Floating Exchange Rates

Disadvantages of fixed exchange rates

- **Conflicts with domestic needs:** Exchange rate targets may limit needed policy actions (e.g., interest rate cuts).
- **Reduced investment & growth:** High interest rates used to defend fixed rates can hurt domestic investment and growth.
- **Risk of recession:** Contractionary policies used to fix exchange rates can cause unemployment and economic downturns.

Fixed versus Floating Exchange Rates

Disadvantages of fixed exchange rates

- **Liquidity issues:** Risk of not having enough or having too much international liquidity (reserves of tradable currencies).
- **Inflexibility to shocks:** Inability to quickly adjust to external shocks like oil price hikes.
- **Speculation risk:** If confidence in the fixed rate is lost, speculation can force damaging devaluations

Fixed versus Floating Exchange Rates

Advantages of free-floating rates

- **Automatic correction:** Exchange rates adjust to eliminate trade imbalances.
- **No reserve needs:** No requirement for central banks to hold large currency reserves.
- **Shock insulation:** More flexibility in responding to global economic fluctuations.
- **Policy independence:** Governments can freely set domestic economic policies, like interest rates.

Fixed versus Floating Exchange Rates

Disadvantages of free-floating rates

- **Volatility:** Exchange rates can fluctuate unpredictably, especially in the short term.
- **Speculation:** Large, rapid capital movements can destabilize exchange rates.
- **Uncertainty:** Fluctuations make it hard for firms to plan long-term trade and investment.
 - Forward contracts help in the short term, but don't eliminate long-term uncertainty.
- **Reduced investment:** Exchange rate risk may discourage foreign and domestic investment.
- **Lack of fiscal discipline:** Governments may pursue inflationary policies without external constraints.

Fixed versus Floating Exchange Rates

- Most countries use **managed floating** systems today.
- **Bretton Woods system (1945–1971)**: Fixed exchange rates pegged to the US dollar, with periodic adjustments.
- **Collapse of Bretton Woods**: Due to global inflation and speculative pressures in the late 1960s.
- **Managed flexibility**: Post-1971 era where central banks occasionally intervene to limit volatility.
- **European attempts at stability**:
 - **ERM (1979)**: Exchange rate bands to prepare for eventual currency union.
 - **Euro introduced (1999)**: Eliminated exchange rate fluctuations within Eurozone countries

Questions